

A Project Report
On
Explainable AI for Clinical Decision Support in Heart
Disease Prediction

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**SCHOOL OF COMPUTING SCIENCE AND ENGINEERING
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CANDIDATE'S DECLARATION

I/We hereby certify that the work which is being presented in the project, entitled “**Explainable AI for Clinical Decision Support in Heart Disease Prediction**” in partial fulfillment of the requirements for the award of the B. Tech. (Computer Science and Engineering) submitted in the School of Computing Science and Engineering of Galgotias University, Greater Noida, is an original work carried out during the period of Aug, 2025 to Jun 2026, under the supervision of Prof **Dr. Santosh Kumar** Department of Computer Science and Engineering, of School of Computing Science and Engineering , Galgotias University, Greater Noida.

The matter presented in the thesis/project/dissertation has not been submitted by me/us for the award of any other degree of this or any other places.

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This is to certify that the above statement made by the candidates is correct to the best of my knowledge.

Dr. Santosh Kumar

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CERTIFICATE

This is to certify that Project Report entitled “**Explainable AI for Clinical Decision Support in Heart Disease Prediction**” which is submitted by Avi Jain, Durgesh Kumar Shukla in partial fulfillment of the requirement for the award of degree B.Tech. in Department of Computer Science and Engineering, Galgotias University, Greater Noida, India is a record of the candidate own work carried out by him/them under my supervision. The matter embodied in this thesis is original and has not been submitted for the award of any other degree

Signature of Examiner(s)

Signature of Supervisor(s)

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Signature of Program Chair

Date:

Place: Greater Noida

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ABSTRACT

Cardiovascular diseases are among the leading causes of death globally, emphasizing the need for timely and accurate diagnosis. In the medical field, Artificial Intelligence (AI) has shown immense potential to assist healthcare professionals in predicting and preventing such conditions. However, most AI-based systems operate as “black boxes,” providing predictions without explaining the underlying reasoning. This lack of transparency often limits their acceptance in clinical settings.

This project, titled “**Explainable AI for Clinical Decision Support in Heart Disease Prediction,**” aims to bridge this gap by combining predictive modeling with interpretability. The system utilizes a **Random Forest classifier** trained on the **UCI Cleveland Heart Disease dataset** to predict the likelihood of a patient having heart disease based on clinical parameters such as age, cholesterol level, resting blood pressure, and chest pain type.

To ensure transparency, the project integrates **SHAP (SHapley Additive exPlanations)**, an explainability technique that visualizes how each feature influences the model’s decision. The user-friendly **Flask web interface** allows clinicians or users to input patient data, receive predictions (High/Low risk), and view SHAP plots showing feature contributions

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CHAPTER-1

INTRODUCTION

1.1 Problem Introduction

Heart disease remains one of the leading causes of mortality worldwide. Traditional diagnostic methods often rely on manual interpretation of various health indicators, which may be time-consuming and prone to human error. **Artificial Intelligence (AI)** can assist by identifying complex patterns in patient data to predict disease risk. However, black-box AI models often lack transparency, making them difficult to trust in medical contexts. Hence, there is a pressing need **for Explainable AI (XAI) systems** that can not only make accurate predictions but also justify them in understandable terms for clinicians.

1.1.2 Project Objective

The main objective of this project is to develop a prototype that:

- Predicts the likelihood of heart disease using patient data.
- Provides interpretable explanations of model predictions using SHAP.
- Offers an easy-to-use web interface for clinical decision support.
- Demonstrates the integration of explainable AI techniques into healthcare workflows.

1.1.3 Scope of the Objective

The project focuses on developing a proof-of-concept prototype for heart disease prediction. It uses the UCI Cleveland Heart Disease dataset for model training and testing. The web application, built with Flask, allows users to input parameters and visualize both predictions and their SHAP-based explanations. The system is designed to be extendable to real-world healthcare data and can support further model and UI enhancements.

CHAPTER 2

REQUIREMENTS SPECIFICATION

2.1 Product Perspective

The proposed system is a standalone, research-level web application intended to serve as a decision-support tool for healthcare professionals, specifically cardiologists and general practitioners. It is not a diagnostic tool; rather, it provides supplementary, evidence-based information to aid the clinician's decision-making process. The system operates "offline" by using a pre-trained ML model and does not require a live internet connection for its core logic, though it is delivered via a web interface.

2.1.1 System Interfaces

User Interface (UI): The primary interface is a Graphical User Interface (GUI) rendered in a standard web browser.

- **Input Page:** A secure web form where the clinician can manually enter a patient's data (e.g., age, sex, cholesterol, resting blood pressure, etc.).
- **Output/Results Page:** A dashboard that appears after submitting the data, displaying the prediction and explanation.

2.1.2 Interfaces

(This section is often interpreted as external data or hardware interfaces. Given 2.1.1 and 2.1.3, this section will define the data and model interfaces.)

- **Data Interface:** The system interfaces with a static, pre-processed dataset (e.g., data.csv) during the training phase. During the inference phase (live prediction), it only interfaces with the data provided by the user via the web form.
- **Model Interface:** The web application's backend interfaces with the saved, serialized ML model (e.g., a model.pkl file) and the corresponding XAI explainer.

2.1.3 Software Interfaces

- **Frontend:** HTML templates rendered using Flask's Jinja2.
- **Backend:** Flask API integrating Python scripts and the trained model.
- **Visualization:** Matplotlib used for SHAP value plots.

2.2 Product Functions

- Accepts user input of 13 medical parameters.
- Predicts heart disease risk (Low/High).
- Displays prediction probability.
- Generates SHAP bar plot showing feature impact on prediction.

2.3 User Characteristics

Intended users include medical practitioners, researchers, and students exploring clinical AI applications. Users are expected to have basic knowledge of health parameters.

2.4 Constraints

- Model trained on a limited dataset (UCI Cleveland).
- Not certified for real medical use.
- Requires numerical input formatting.
- SHAP visualization may take extra computation time.

2.5 Assumptions and Dependencies

- Dataset is accurate and representative.
- Users input valid numeric values.
- Dependencies include Flask, scikit-learn, SHAP, matplotlib, and joblib.

CHAPTER 3

SYSTEM DESIGN

3.1 System Architecture Overview

The architecture of the project “Explainable AI for Clinical Decision Support in Heart Disease Prediction” is designed in a modular three-tier structure, comprising the Data Layer, Application Logic Layer, and Presentation Layer. Each layer performs a distinct function, ensuring modularity, scalability, and maintainability of the system.

At its core, the system integrates a machine learning model for disease prediction with an explainability module (SHAP) for model interpretation. The backend communicates with the trained model and explainability framework, while the frontend interface enables end-users (clinicians or students) to interact with the system through a web browser.

3.2 Architectural Layers

1. Data Layer

This layer handles all data-related operations including dataset acquisition, preprocessing, and model training.

- The system uses the UCI Cleveland Heart Disease Dataset, containing medical features such as age, sex, chest pain type, resting BP, cholesterol, and more.
- Missing values (denoted by '?') are handled using pandas for data cleaning.
- The dataset is split into training and testing subsets (80–20 ratio).
- A Random Forest Classifier is trained to learn patterns and relationships between clinical features and heart disease occurrence.
- The trained model and metadata (feature names) are stored in a serialized file `model.joblib` using `joblib`, allowing the Flask app to load it efficiently during inference.

Key Components:

- `trainmodel.py` script (handles model training)
 - `heart.csv` dataset file
 - `model.joblib` (stored model for deployment)
-

2. Application Logic Layer

This layer acts as the core processing engine of the application. It is implemented using Flask, a lightweight Python web framework.

- The Flask app (app.py) manages HTTP requests, data validation, and interaction with the trained model.
- When the user submits patient information via the web form, the app performs the following steps:
 1. Input Processing: Retrieves 13 numerical input values corresponding to medical features.
 2. Prediction: Loads the trained model and computes the probability of heart disease using `model.predict_proba()`.
 3. Explainability: Invokes SHAP's TreeExplainer to compute Shapley values for the given input, quantifying each feature's contribution to the prediction.
 4. Visualization: Uses matplotlib to generate a SHAP bar chart highlighting the most influential features in descending order of importance.
 5. Response Rendering: The computed prediction, probability, and SHAP image are returned to the frontend for display.

This layer essentially bridges the AI model and the web interface, ensuring smooth data flow and interpretability output generation.

Key Components:

- Flask routes (/ endpoint)
 - SHAP TreeExplainer integration
 - Matplotlib visualizations
 - Dynamic rendering with Jinja2 templates
-

3. Presentation Layer

This layer serves as the user interface, allowing users to interact with the AI model intuitively.

- The HTML frontend (template index.html) displays an input form with labeled fields for all 13 medical attributes.
- Upon submission, the page displays:
 - Prediction result: “High Risk” or “Low Risk”

- Probability score: Likelihood of heart disease (e.g., 0.784)
- SHAP Bar Chart: Explains which features contributed positively or negatively to the model's decision.
- The interface is designed for simplicity and usability, suitable for quick clinical assessments or educational demonstrations.

Key Components

- HTML + Jinja2 templates for rendering results
- Static folder for SHAP images (static/shap.png)
- CSS (optional styling for readability)

CHAPTER 4

IMPLEMENTATION

4.1 Introduction

This chapter details the system implementation, including model training, backend logic, and web interface integration.

4.2 System Implementation Overview

- The model is trained using `trainmodel.py` with a Random Forest Classifier.
- After training, the model is serialized as `model.joblib`.
- The Flask app (`app.py`) loads the model, takes user input, computes predictions, and generates SHAP visualizations.

4.3 Tools and Technologies Used

Category	Tools/Frameworks
Programming Language	Python
Framework	Flask
Libraries	scikit-learn, SHAP, matplotlib, pandas, numpy
Model	Random Forest Classifier
Dataset	UCI Cleveland Heart Disease Dataset
Deployment	Localhost Flask Server

4.4 Implementation Steps

1. Download and preprocess dataset.
2. Train Random Forest model using scikit-learn.
3. Save model using joblib.
4. Build Flask web interface for user input.
5. Integrate SHAP explainability and visualize feature impact.
6. Test prototype with sample data.

4.5 Result and Performance Analysis

The Random Forest classifier achieved an accuracy of approximately **85–88%** on test data. SHAP visualizations successfully highlighted key predictors such as **chest pain type**, **cholesterol**, **resting BP**, and **maximum heart rate achieved**, allowing transparent interpretation of the model's reasoning.

CHAPTER 5

CONCLUSION

5.1 Conclusion and Future Scope

The developed system demonstrates the potential of **Explainable AI in clinical decision support**. By combining predictive modeling with interpretability, clinicians can make informed and transparent diagnostic decisions. Future enhancements could include:

- Integration with real-world hospital EHR data.
- Use of deep learning models with integrated explainability.
- Deployment on cloud platforms for scalability.
- Improved visualization dashboards for clinicians.